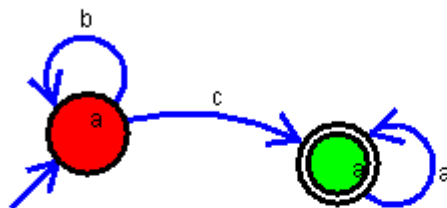


<http://www.cburch.com/proj/autosim/download.html>

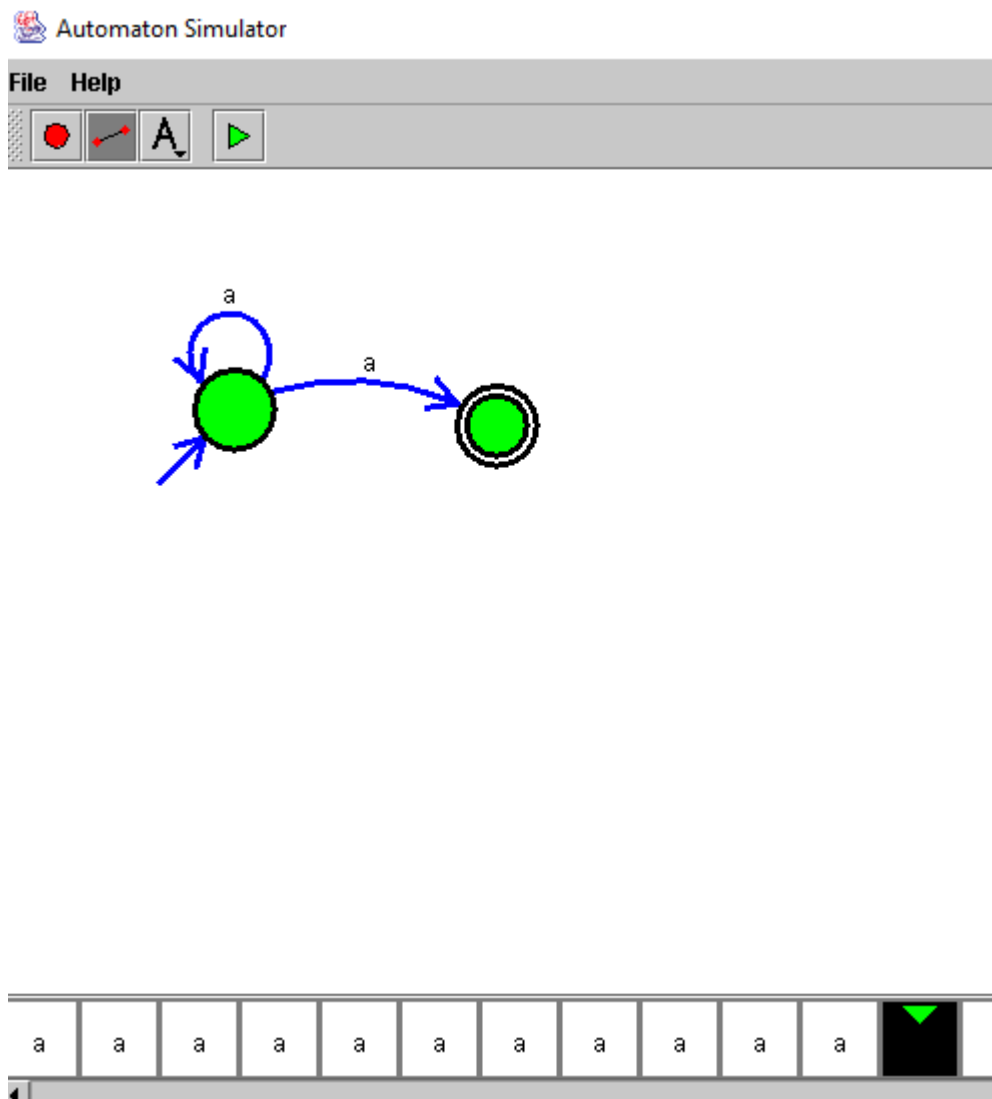
1. Design DFA to accept bcaaaaaaaaaaaaaa, bc, and c

 Automaton Simulator



b	c	a	a	a	a	a	a	a	a		
---	---	---	---	---	---	---	---	---	---	---	--

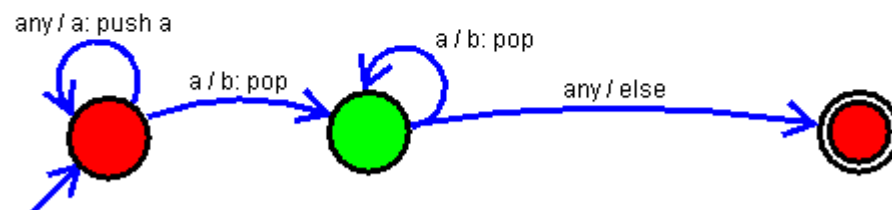
2.Design NFA to accept aaaaaa



3.Design PDA for the input a^nb^n

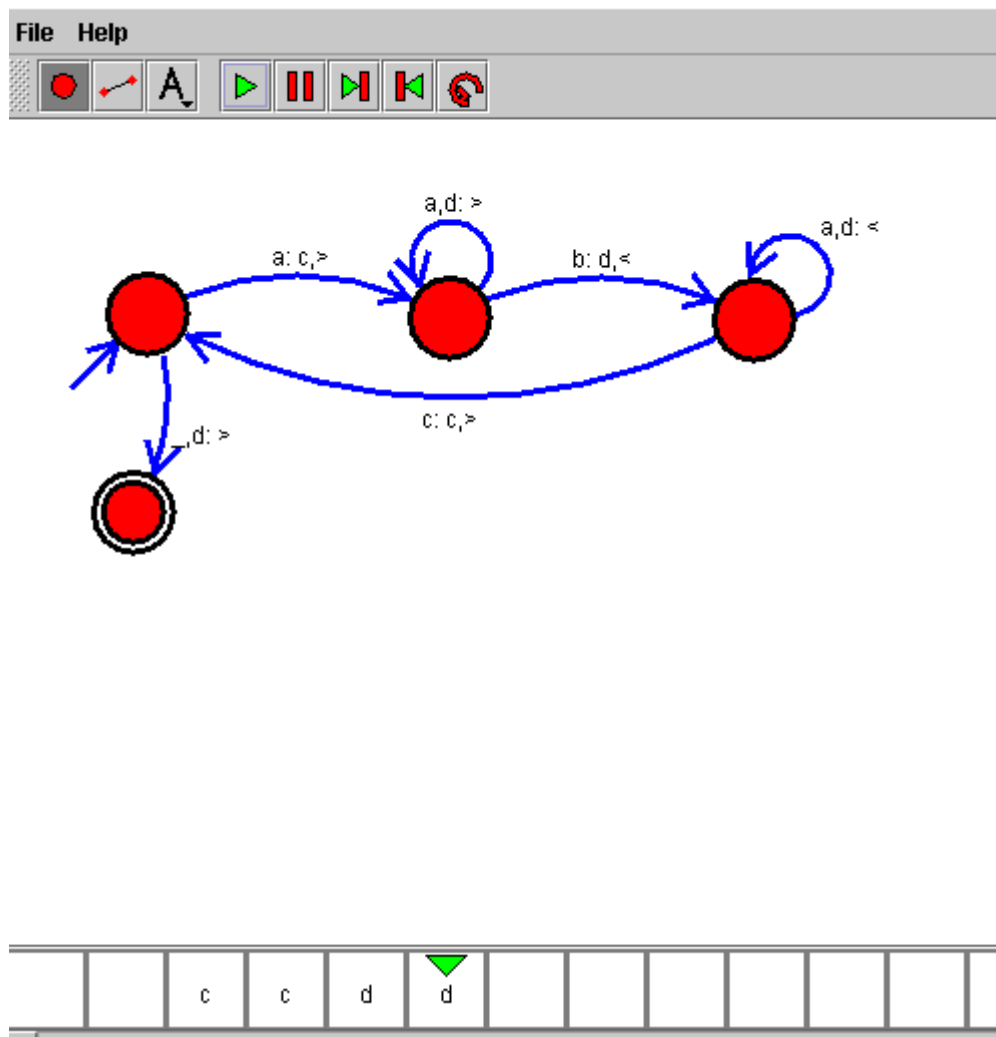
Automaton Simulator

File Help

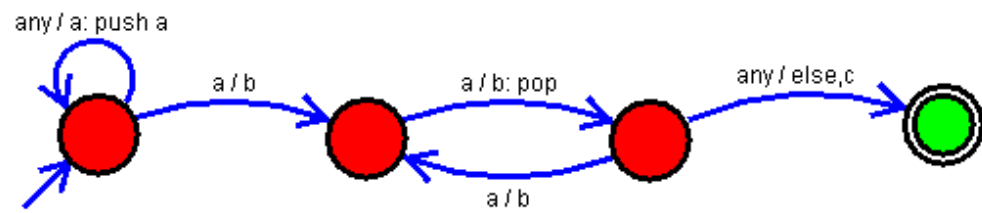


4.Design Tm For input a^nb^n

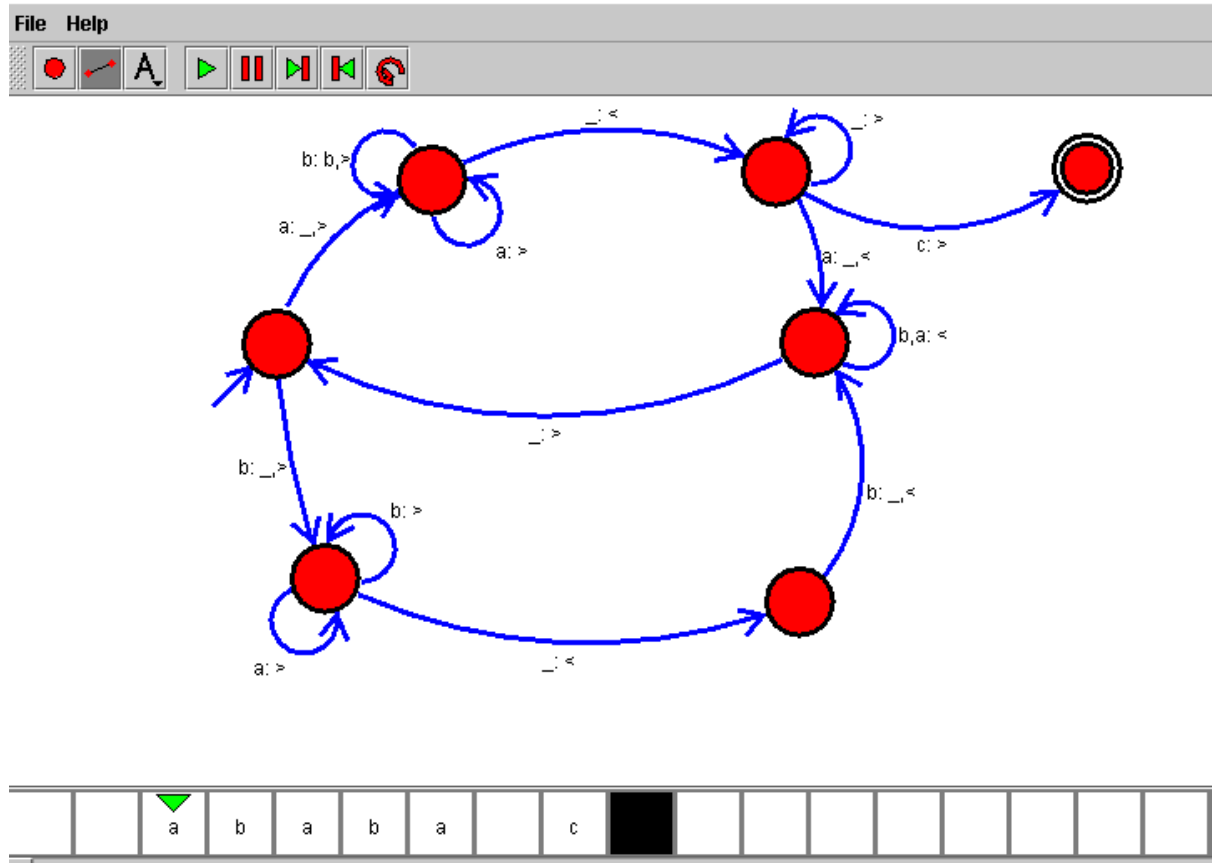
 Automaton Simulator



5 .Design PDA for input aabbbbc ($L=a^nb^{2n}$)



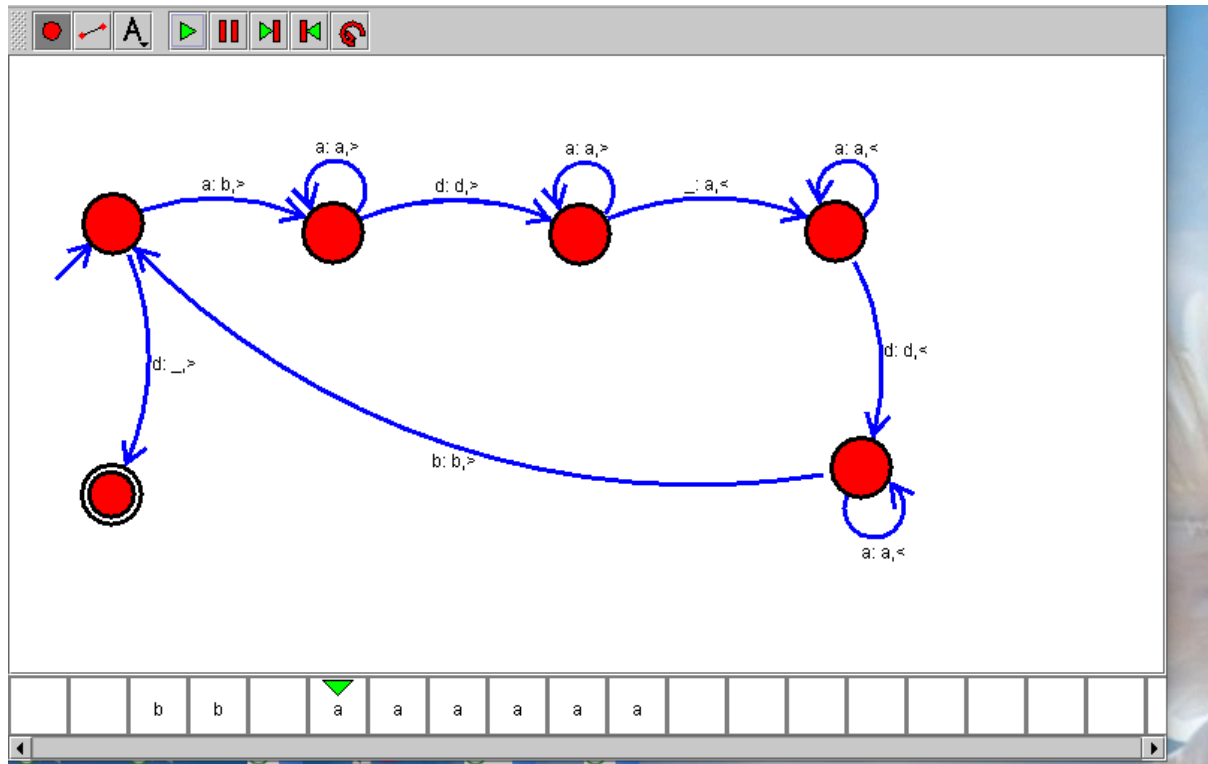
6.TM Simulation for Palindrome W= ababa c



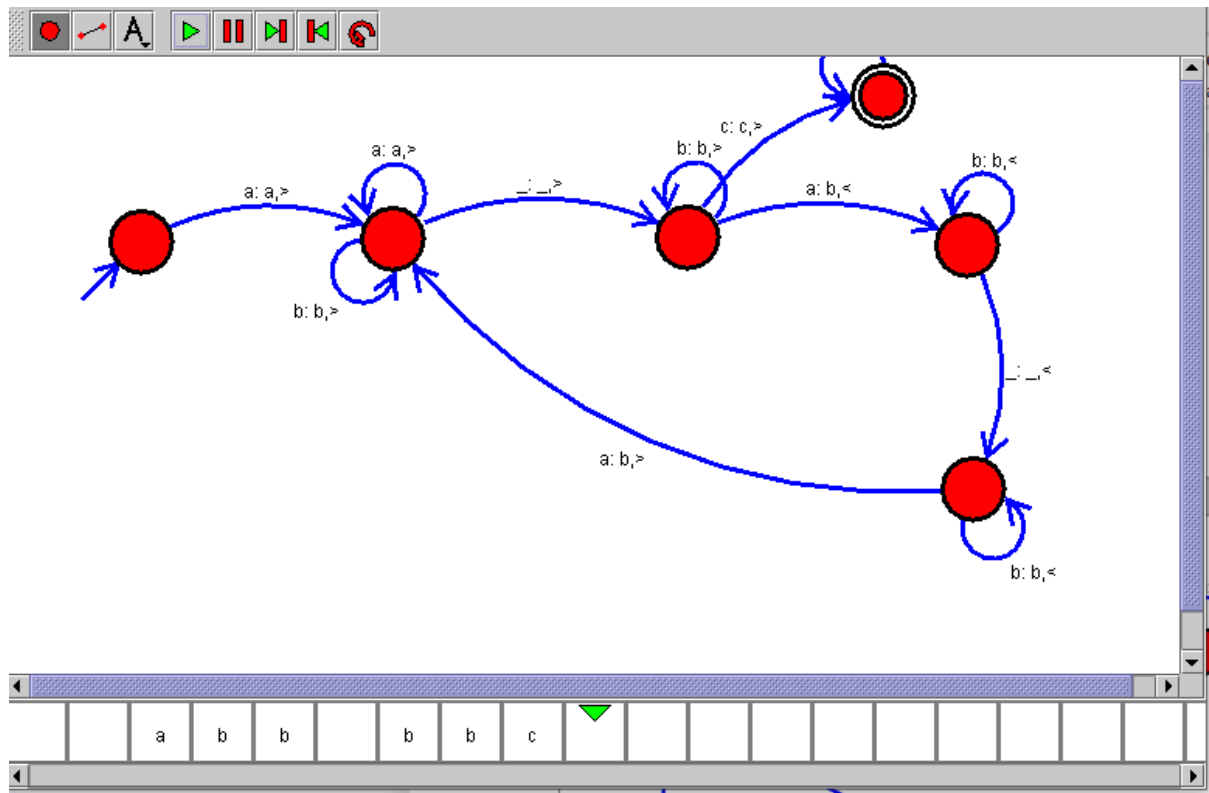
7.Design TM to perform addition of following

W= aa + aaaa

After Addition of a's = aaaaaa

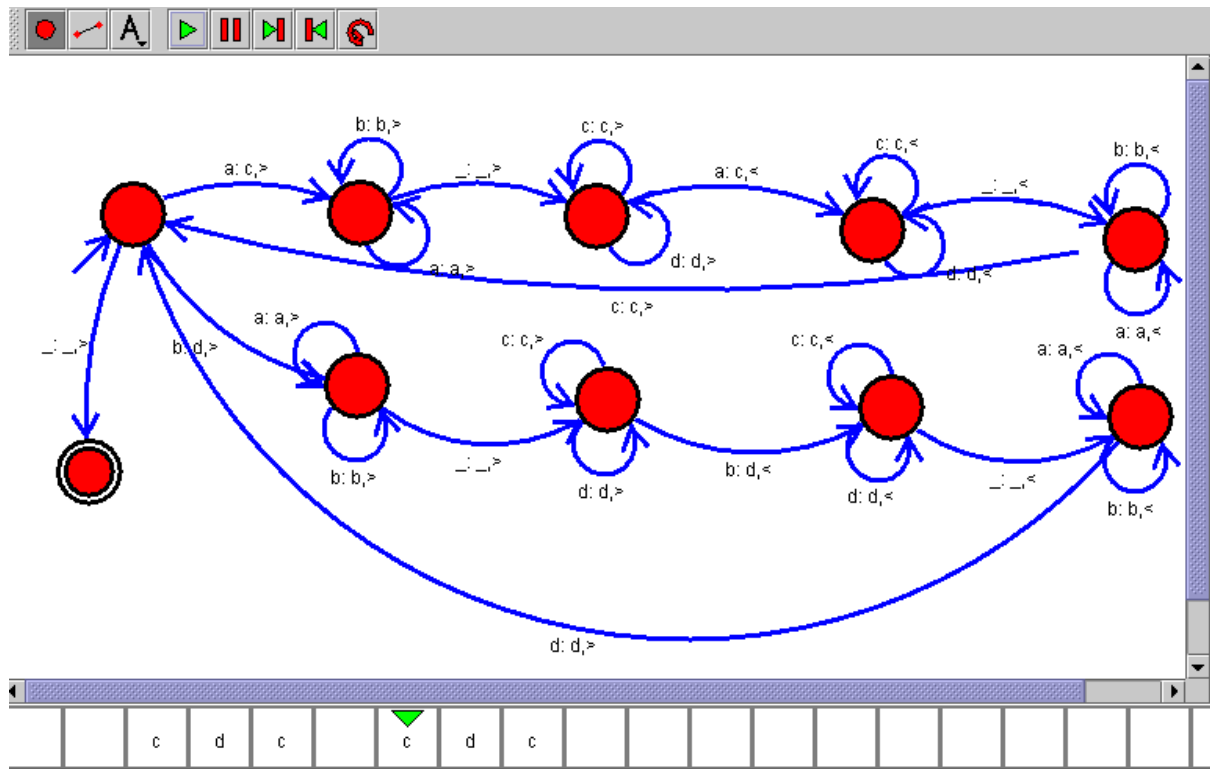


The Result of Subtraction is = a



9.Design TM to perform string comparison

W = aba aba



10. Write CFG to product string which consists of substring 'aa'

Create

Input your context-free grammar (CFG) here. The start symbol has already been filled in for you.

- The left-hand nonterminal of each production must be filled in.
- [ϵ] - An empty text field corresponds to epsilon.
- [|] - For "or", use the standard pipe character that you use while coding.
- Input is case-sensitive. Whitespace is not ignored.

Reset

Example

S → AaaA

A → aA | bA | ϵ ⊗

Test

To test the CFG above, input test strings here, one per line. An empty line corresponds to the empty string. Results will be shown automatically.

aaab

Test Results for CFG

#	String	Matches	
1	"aaab"	Yes (ambiguously)	Derivation One Derivation Two
Rule		Application	Result
Start → S		Start	S
S → AaaA		S	AaaA
A → aA		AaaA	aAaaA
A → ϵ		aAaaA	aaaA
A → bA		aaaA	aaabA
A → ϵ		aaabA	aaab

11. Write Context Free Grammar to Generate Palindrome

Verify

This is the CFG you have input above:

Start symbol: **S**

S $\rightarrow \epsilon \mid aSa \mid bSb \mid a \mid b$

Some strings from the language of this grammar:

```
abba
bbbb
abababbbbbbababa
-
```

12.

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Automaton Simulator: DFA NFA PDA

Test / Debug:

Bulk Testing →

Accept (one per line):
AAAAAB

Diagram: start (start) → B → s0 (accept)

Set Transition Character

Empty transitions not allowed for DFAs
Read from Input

start → B → s0

Cancel Delete Save

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Automaton Simulator: DFA NFA PDA Examples

Test / Debug:

Bulk Testing →

Accept (one per line):
AACBB

Reject (one per line):

Diagram: start (start) → C,ε,ε → s0 (accept) → ε,ε,ε → s1 (accept)

Set Transition Characters

Blank for Empty String: ε
Read from Input | Pop off Stack | Push on Stack

s0 → A → s1

Cancel Delete Save

Test Results:
Accept: AACBB -- Pass
Reject: [Empty String] -- Fail

13.



est / Debug:

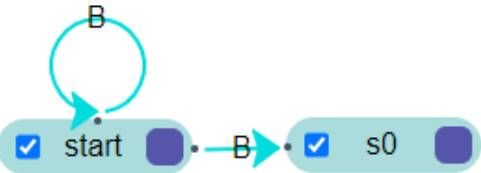


Bulk Testing 

Accept (one per line):
BBBBBBBBBBBB

Reject (one per line):

Automaton Simulator:



```
graph LR; start((start)) -- B --> start; start -- B --> s0((s0));
```

Test Results:
Accept: BBBBBBBBBBBBBB -- Pass

14. Write CFG which will produce string over set = {a,b} that start with 'a' and end with 'b'

Verify

This is the CFG you have input above:

Start symbol: **S**

$S \rightarrow \epsilon \mid aAb$

$A \rightarrow aA \mid bA \mid \epsilon$

Some strings from the language of this grammar:

```
aabbaab
abb
aaabaabbaababab
abbb
aabaaaaab
abaababb
abaabbaab
ε
abbbab
aabbababbabb
```

15. Design PDA to accept event number a 's and even number b 's